Kali Linux Setup & OWASP Juice Shop Installation Guide

Part 1: Kali Linux Setup Options

Option 1: Setup Using Hyper-V

Prerequisites:

- Windows 10 Pro, Enterprise, or Education (or Windows 11)
- Virtualization enabled in BIOS
- Minimum 4GB RAM (8GB+ recommended)
- 20GB+ free disk space

Steps:

1. **Enable Hyper-V**
   - Open PowerShell as Administrator
   - Run: `Enable-WindowsOptionalFeature -Online -FeatureName Microsoft-Hyper-V -All`
   - Restart your computer when prompted
2. **Download Kali Linux**
   - Visit: https://www.kali.org/get-kali/
   - Download the Hyper-V image (Pre-built VM)
   - Extract the downloaded .7z file
3. **Import Kali VM into Hyper-V**
   - Open Hyper-V Manager
   - Click "Import Virtual Machine" from the Actions menu
   - Browse to the extracted Kali folder
   - Select the folder containing the VM files
   - Choose "Copy the virtual machine" option
   - Complete the import wizard
4. **Configure VM Settings**
   - Right-click the imported VM → Settings
   - Adjust RAM (minimum 2GB, recommended 4GB+)

- Configure network adapter (Default Switch or External)
- Apply changes

5. **Start and Login**
   - Start the VM
   - Default credentials: `kali` / `kali`
   - Change password immediately after first login

Option 2: Setup Using Vagrant

Prerequisites:

- Vagrant installed (https://www.vagrantup.com/downloads)
- VirtualBox or VMware installed
- Command-line access

Steps:

1. **Create Project Directory**

```bash
mkdir kali-vagrant
cd kali-vagrant
```

2. **Initialize Vagrant with Kali**

```bash
vagrant init kalilinux/rolling
```

3. **Customize Vagrantfile (Optional)**
   - Open `Vagrantfile` in a text editor
   - Adjust settings as needed:

```ruby
```

```
Vagrant.configure("2") do |config|
  config.vm.box = "kalilinux/rolling"
  config.vm.provider "virtualbox" do |vb|
    vb.memory = "4096"
    vb.cpus = 2
    vb.gui = true
  end
end
```

4. **Start Vagrant VM**

```bash
vagrant up
```

5. **Access the VM**

```bash
# SSH access
vagrant ssh

# Or use the GUI if enabled
```

6. **Useful Vagrant Commands**

```bash
vagrant halt        # Stop the VM
vagrant suspend     # Suspend the VM
vagrant resume      # Resume suspended VM
vagrant destroy     # Delete the VM
vagrant reload      # Restart VM with new config
```

Option 3: Setup Using Docker

Prerequisites:

- Docker installed (https://docs.docker.com/get-docker/)
- Basic Docker knowledge

Note: Docker provides a lightweight Kali environment but with limitations compared to full VMs.

Steps:

1. **Pull Official Kali Docker Image**

```bash
docker pull kalilinux/kali-rolling
```

2. **Run Kali Container (Basic)**

```bash
docker run -ti kalilinux/kali-rolling /bin/bash
```

3. **Run with Persistent Storage**

```bash
# Create a volume for persistent data
docker volume create kali-data

# Run with mounted volume
docker run -ti -v kali-data:/root kalilinux/kali-rolling /bin/bash
```

4. **Run with Network Tools (Recommended)**

```bash
docker run -ti --privileged --network host \
  -v kali-data:/root \
  --name kali-container \
  kalilinux/kali-rolling /bin/bash
```

5. **Create a Custom Kali Image (Optional)**
   - Create a `Dockerfile`:

```dockerfile
```

```
FROM kalilinux/kali-rolling
RUN apt update && apt install -y \
    kali-linux-default \
    kali-tools-top10
```

- Build the image:

```bash
docker build -t my-kali .
docker run -ti my-kali /bin/bash
```

6. **Manage Docker Container**

```bash
# Start stopped container
docker start -ai kali-container

# Stop container
docker stop kali-container

# Remove container
docker rm kali-container
```

Part 2: Starting Kali Linux & Opening the Terminal

For First-Time Users: What is a Terminal?

A terminal (also called command line or console) is a text-based interface where you type commands to interact with your computer. Think of it as an alternative to clicking icons and menus - instead, you type instructions.

Option 1: Starting Kali in Hyper-V

Step 1: Open Hyper-V Manager

- Press the Windows key and type "Hyper-V"
- Click on "Hyper-V Manager"

Step 2: Start Your Kali VM

- In Hyper-V Manager, you'll see your Kali virtual machine in the list
- Right-click on the Kali VM
- Click "Connect" (this opens a window showing the VM)
- Click "Start" (green play button at the top)

Step 3: Log In to Kali

- Wait for Kali to boot up (you'll see a login screen)
- Username: `kali`
- Password: `kali` (you won't see characters as you type - this is normal for security)
- Press Enter

Step 4: Open the Terminal Once logged in, you'll see the Kali desktop. Here are three ways to open the terminal:

Method 1 (Easiest):

- Look at the top-left of the screen
- Click the terminal icon (looks like a black window with `>_` symbol)

Method 2:

- Click the "Applications" menu in the top-left corner
- Hover over "System Tools"
- Click "Terminal"

Method 3:

- Right-click anywhere on the desktop
- Select "Open Terminal Here"

Option 2: Starting Kali in Vagrant

Step 1: Open Your Command Prompt or Terminal

Windows:

- Press Windows key + R

- Type `cmd` and press Enter
- OR press Windows key, type "Command Prompt", and click it

Mac/Linux:

- Press Command + Space (Mac) or Ctrl + Alt + T (Linux)
- Type "terminal" and press Enter

Step 2: Navigate to Your Kali Directory

```bash
cd kali-vagrant
```

(This moves you into the folder where you set up Vagrant)

Step 3: Start Vagrant

```bash
vagrant up
```

(Wait for it to start - this may take a few minutes)

Step 4: Access Kali

If you enabled GUI in Vagrantfile:

- VirtualBox window will open automatically showing Kali desktop
- Log in with username: `vagrant` password: `vagrant`
- Open terminal using the methods described in the Hyper-V section above

If using SSH only (no GUI):

```bash
vagrant ssh
```

- You're now in the Kali terminal automatically!
- No GUI, just text commands

Option 3: Starting Kali in Docker

Step 1: Open Your Terminal/Command Prompt

- Follow the same steps as Vagrant Option above for your operating system

Step 2: Start Your Kali Container

If you created a named container:

```bash
docker start -ai kali-container
```

If starting fresh:

```bash
docker run -ti --privileged --network host kalilinux/kali-rolling /bin/bash
```

You're now in the Kali terminal!

- Docker Kali runs in terminal-only mode (no graphical interface)
- You'll see a prompt that looks like: `root@xxxxxx:/#`

Understanding the Terminal

What you'll see:

When you open a terminal, you'll see something like:

```
┌──(kali㉿kali)-[~]
└─$
```

OR

```
kali@kali:~$
```

This is called the "command prompt" - it's waiting for you to type a command.

Parts of the prompt:

- `kali@kali` - your username and computer name
- `~` - your current location (~ means home directory)
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Part 1: Kali Linux Setup Options

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   - Complete the import wizard
4. **Configure VM Settings**

- Right-click the imported VM → Settings
- Adjust RAM (minimum 2GB, recommended 4GB+)
- Configure network adapter (Default Switch or External)
- Apply changes

5. **Start and Login**
   - Start the VM
   - Default credentials: `kali` / `kali`
   - Change password immediately after first login

Option 2: Setup Using Vagrant

Prerequisites:

- Vagrant installed (https://www.vagrantup.com/downloads)
- VirtualBox or VMware installed
- Command-line access

Steps:

1. **Create Project Directory**

```bash
mkdir kali-vagrant
cd kali-vagrant
```

2. **Initialize Vagrant with Kali**

```bash
vagrant init kalilinux/rolling
```

3. **Customize Vagrantfile (Optional)**
   - Open `Vagrantfile` in a text editor
   - Adjust settings as needed:

```ruby
```

```
Vagrant.configure("2") do |config|
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  config.vm.provider "virtualbox" do |vb|
    vb.memory = "4096"
    vb.cpus = 2
    vb.gui = true
  end
end
```

4. **Start Vagrant VM**

```bash
vagrant up
```

5. **Access the VM**

```bash
# SSH access
vagrant ssh

# Or use the GUI if enabled
```

6. **Useful Vagrant Commands**

```bash
vagrant halt        # Stop the VM
vagrant suspend     # Suspend the VM
vagrant resume      # Resume suspended VM
vagrant destroy     # Delete the VM
vagrant reload      # Restart VM with new config
```

Option 3: Setup Using Docker

Prerequisites:

- Docker installed (https://docs.docker.com/get-docker/)
- Basic Docker knowledge

Note: Docker provides a lightweight Kali environment but with limitations compared to full VMs.

Steps:

1. **Pull Official Kali Docker Image**

```bash
docker pull kalilinux/kali-rolling
```

2. **Run Kali Container (Basic)**

```bash
docker run -ti kalilinux/kali-rolling /bin/bash
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3. **Run with Persistent Storage**

```bash
# Create a volume for persistent data
docker volume create kali-data

# Run with mounted volume
docker run -ti -v kali-data:/root kalilinux/kali-rolling /bin/bash
```

4. **Run with Network Tools (Recommended)**

```bash
docker run -ti --privileged --network host \
  -v kali-data:/root \
  --name kali-container \
  kalilinux/kali-rolling /bin/bash
```

5. **Create a Custom Kali Image (Optional)**
   - Create a `Dockerfile`:

```dockerfile
```

```
FROM kalilinux/kali-rolling
RUN apt update && apt install -y \
    kali-linux-default \
    kali-tools-top10
```

- Build the image:

```bash
docker build -t my-kali .
docker run -ti my-kali /bin/bash
```

6. **Manage Docker Container**

```bash
# Start stopped container
docker start -ai kali-container

# Stop container
docker stop kali-container

# Remove container
docker rm kali-container
```

- indicates you're a regular user (`#` means you're root/admin)

Basic Terminal Tips:

1. **Type carefully** - Spelling matters! `Update` is different from `update`
2. **Copy and Paste:**
   - **Copy:** Ctrl + Shift + C (in terminal) or select text and right-click → Copy
   - **Paste:** Ctrl + Shift + V (in terminal) or right-click → Paste
3. **Cancel a command:**
   - Press `Ctrl + C` to stop a running command
4. **Clear the screen:**
   - Type `clear` and press Enter
5. **Command history:**
   - Press ↑ (up arrow) to see previous commands

- Press ↓ (down arrow) to go forward through history

6. **Auto-complete:**
   - Type the first few letters and press Tab to auto-complete file/folder names

---

Basic Navigation Commands (Try These!)

```bash
# See where you are
pwd

# List files in current folder
ls

# List files with details
ls -la

# Go to home directory
cd ~

# Go to a specific folder
cd Downloads

# Go back one folder
cd ..

# Create a new folder
mkdir my-folder

# See what's running
top
# (Press 'q' to quit)
```

Testing Your Terminal

Once your terminal is open, try typing this command and press Enter:

```bash
whoami
```

You should see your username (probably `kali`). If you see this, your terminal is working!

Try another:

```bash
ls
```

This lists the files in your current location.

Congratulations! You're now ready to use the terminal for the OWASP Juice Shop installation below.

Part 3: Post-Installation Setup & OWASP Juice Shop

Prerequisites

- Kali Linux installed and running (using any method above)
- Terminal open (see Part 2 above)
- Internet connection
- Root or sudo access

Step 1: Update Kali Linux

Update package lists and upgrade system:

```bash
sudo apt update
sudo apt upgrade -y
sudo apt dist-upgrade -y
```

Clean up unnecessary packages:

```bash
sudo apt autoremove -y
sudo apt autoclean
```

Reboot (if kernel was updated):

```bash
sudo reboot
```

Step 2: Install Required Dependencies

Install Node.js and npm (required for OWASP Juice Shop):

```bash
# Install Node.js from NodeSource repository
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs

# Verify installation
node --version
npm --version
```

Install Git (if not already installed):

```bash
sudo apt install -y git
```

Step 3: Install OWASP Juice Shop

Method A: From Source (Recommended)

1. **Clone the repository:**

```bash
cd ~
git clone https://github.com/juice-shop/juice-shop.git
cd juice-shop
```

2. **Install dependencies:**

```bash
npm install
```

3. **Start Juice Shop:**

```bash
npm start
```

4. **Access the application:**
   - Open browser and navigate to: `http://localhost:3000`

Method B: Using Docker (if Docker is available)

```bash
docker pull bkimminich/juice-shop
docker run -d -p 3000:3000 --name juice-shop bkimminich/juice-shop
```

Access at: `http://localhost:3000`

Method C: Using npm package (Quick Install)

```bash
npm install -g juice-shop
juice-shop
```

Step 4: Running and Managing OWASP Juice Shop

Start Juice Shop (from source directory):

```bash
cd ~/juice-shop
npm start
```

Run in background:

```bash
nohup npm start &
```

Stop Juice Shop:

```bash
# If running in foreground: Press Ctrl+C

# If running in background:
pkill -f "npm start"
# or
ps aux | grep juice-shop
kill [PID]
```

Update Juice Shop to latest version:

```bash
cd ~/juice-shop
git pull
npm install
npm start
```

Step 5: Configure Juice Shop (Optional)

Create custom configuration:

```bash
cd ~/juice-shop
cp config/default.yml config/custom.yml
```

Edit custom configuration:

```bash
nano config/custom.yml
```

Run with custom config:

```bash
NODE_ENV=custom npm start
```

Step 6: Useful Commands

Check if Juice Shop is running:

```bash
netstat -tulpn | grep 3000
# or
lsof -i :3000
```

View logs:

```bash
cd ~/juice-shop
tail -f nohup.out
```

Reset Juice Shop database:

```bash
cd ~/juice-shop
rm data/juiceshop.sqlite
npm start
```

Troubleshooting

Port already in use:

```bash
# Find process using port 3000
sudo lsof -i :3000
# Kill the process
sudo kill -9 [PID]
```

npm installation errors:

```bash
```

```
# Clear npm cache
npm cache clean --force
# Remove node_modules and reinstall
rm -rf node_modules
npm install
```

Permission errors:

```bash
# Fix npm permissions
sudo chown -R $USER:$USER ~/.npm
sudo chown -R $USER:$USER ~/juice-shop
```

Additional Tips

1. **Create a startup script:**

```bash
nano ~/start-juiceshop.sh
```

Add:

```bash
#!/bin/bash
cd ~/juice-shop
npm start
```

Make executable:

```bash
chmod +x ~/start-juiceshop.sh
```

2. **Access from other machines on your network:**
   - Start Juice Shop with host binding:

```bash
```

```
HOST=0.0.0.0 npm start
```

- Access via: `http://[kali-ip]:3000`

3. **Keep Kali and tools updated regularly:**

```bash
sudo apt update && sudo apt upgrade -y
```

Quick Reference Commands

```bash
# System Update
sudo apt update && sudo apt upgrade -y

# Start Juice Shop
cd ~/juice-shop && npm start

# Update Juice Shop
cd ~/juice-shop && git pull && npm install

# Check Juice Shop status
netstat -tulpn | grep 3000

# Stop all node processes
pkill -f node
```